

Clinic Scheduling System

Operational workflow and software design

Preview document generated from the complex MODT testing model.

SYSTEM	ClinicFlow
AUDIENCE	Product and engineering
STATUS	Preview

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1 Clinic Scheduling System

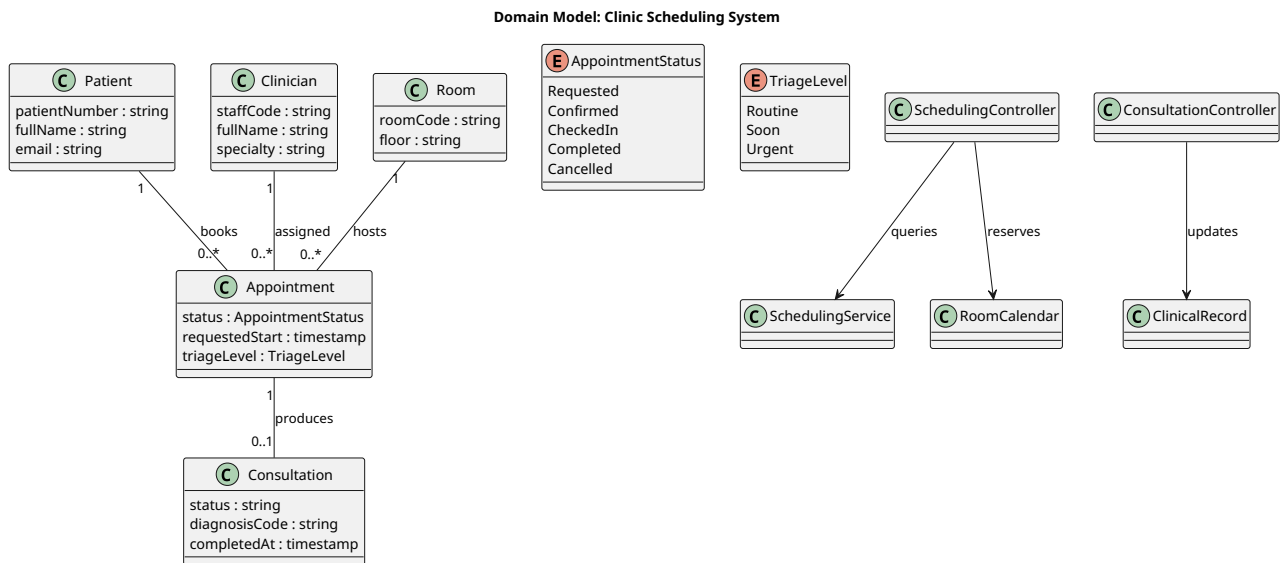
Requirements and design report

Patients book appointments, staff confirm availability, and clinicians complete consultations.

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1.1 Domain

1.1.1 Structural Views



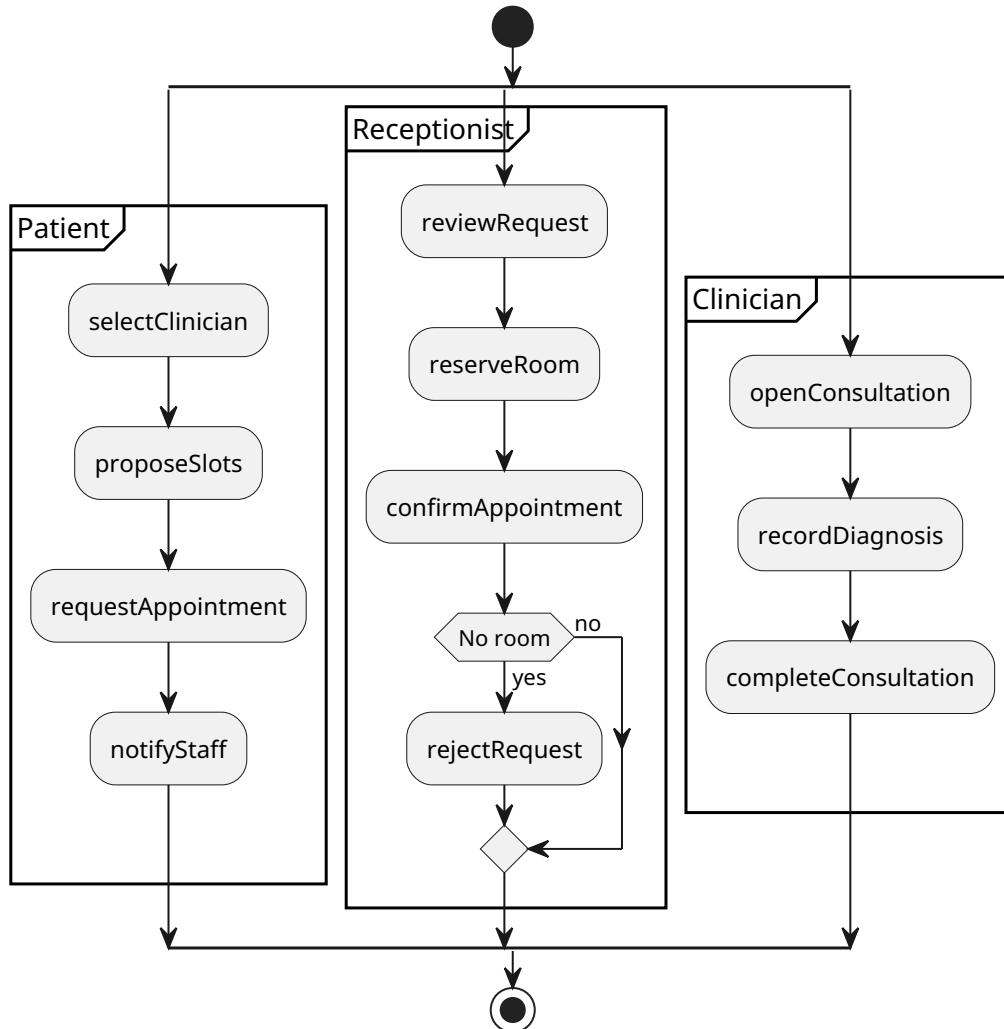
1.1.2 Glossary

Term	Definition	Rules
Appointment	Reserved time slot between a patient and clinician.	Status is Requested, Confirmed, CheckedIn, Completed, or Cancelled.
Consultation	Clinical encounter attached to a confirmed appointment.	A consultation cannot be completed before check-in.
Triage Level	Priority assigned by intake staff.	Allowed values are Routine, Soon, Urgent.

1.2 Analysis

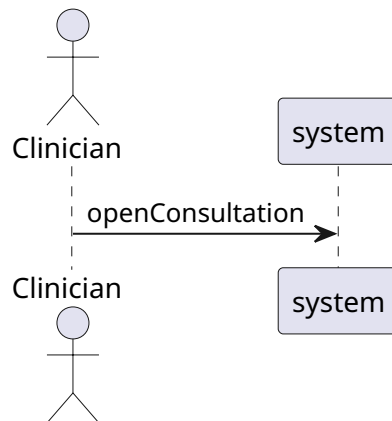
1.2.1 Activity Flows

Activity Diagram: Clinic Scheduling System

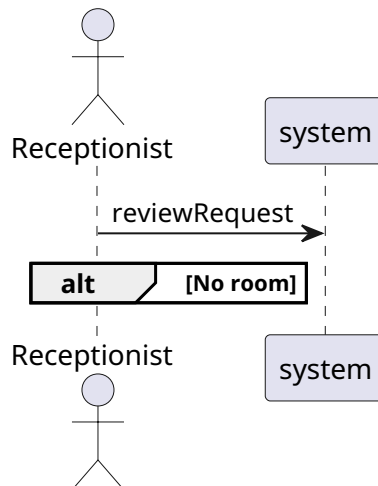


1.2.2 System Interactions

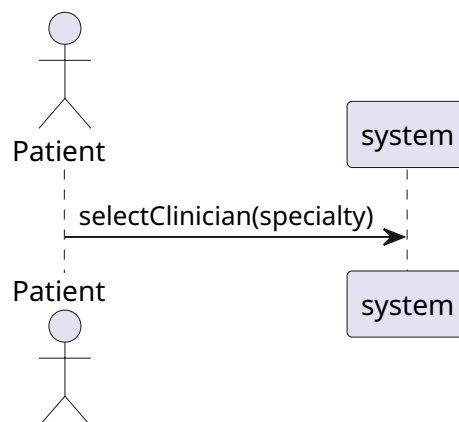
System Sequence Diagram: CompleteConsultation



System Sequence Diagram: ConfirmAppointment



System Sequence Diagram: RequestAppointment



1.2.3 Supplementary Specification

Category	Requirement	Description
usability	Fast booking	A returning patient can request an appointment without re-entering profile data.
reliability	Audit trail	Appointment status changes are recorded.
privacy	Patient data	Clinical notes are visible only to authorized staff.

1.2.4 Use Cases

Use Case	Primary Actor	Steps	Preconditions	Postconditions
RequestAppointment	Patient	4	1	1
ConfirmAppointment	Receptionist	4	1	1
CompleteConsultation	Clinician	3	1	1

1.2.4.1 REQUESTAPPOINTMENT

Description: Patient requests a clinician and time window.

Actor: Patient

Preconditions: - Patient profile exists

Flow of Events: 1. selectClinician - specialty: string 2. proposeSlots (Target: SchedulingService) - clinicianId: string 3. requestAppointment - slotId: string 4. notifyStaff (Target: ReceptionDesk)

Postconditions: - `Appointment.status = Requested`

1.2.4.2 CONFIRMAPPOINTMENT

Description: Receptionist confirms or rejects a requested appointment.

Actor: Receptionist

Preconditions: - `Appointment.status = Requested`

Flow of Events: 1. reviewRequest 2. reserveRoom (Target: RoomCalendar) - appointmentId: string 3. confirmAppointment 4. Alt: No room rejectRequest

Postconditions: - `Appointment.status = Confirmed`

1.2.4.3 COMPLETECONSULTATION

Description: Clinician records notes and completes the consultation.

Actor: Clinician

Preconditions: - `Appointment.status = CheckedIn`

Flow of Events: 1. openConsultation 2. recordDiagnosis (Target: ClinicalRecord) - diagnosisCode: string 3. completeConsultation

Postconditions: - `Consultation.status = Completed`

1.2.5 System Operations

Operation	Actor	Use Case
requestAppointment(slotId: string)	Patient	RequestAppointment
confirmAppointment(appointmentId: string)	Receptionist	ConfirmAppointment
completeConsultation(consultationId: string)	Clinician	CompleteConsultation

1.2.5.1 REQUESTAPPOINTMENT(SLOTID: STRING)

Actor: Patient

Use Case: RequestAppointment

Preconditions: - Patient profile exists

Postconditions: - `Appointment.status = Requested`

1.2.5.2 CONFIRMAPPOINTMENT(APPOINTMENTID: STRING)

Actor: Receptionist

Use Case: ConfirmAppointment

Preconditions: - `Appointment.status = Requested`

Postconditions: - `Appointment.status = Confirmed`

1.2.5.3 COMPLETECONSULTATION(CONSULTATIONID: STRING)

Actor: Clinician

Use Case: CompleteConsultation

Preconditions: - `Appointment.status = CheckedIn`

Postconditions: - `Consultation.status = Completed`

1.2.6 Operation Contracts

1.2.6.1 REQUESTAPPOINTMENT(SLOTID)

Use Case: RequestAppointment

Preconditions: - Patient profile exists

Postconditions: - Appointment instance is created - `Appointment.status = Requested` - Patient rel “-” Appointment is formed

1.2.6.2 CONFIRMAPPOINTMENT(APPOINTMENTID)

Use Case: ConfirmAppointment

Preconditions: - `Appointment.status = Requested`

Postconditions: - `Appointment.status = Confirmed` - Room rel “-” Appointment is formed

1.2.6.3 COMPLETECONSULTATION(CONSULTATIONID)

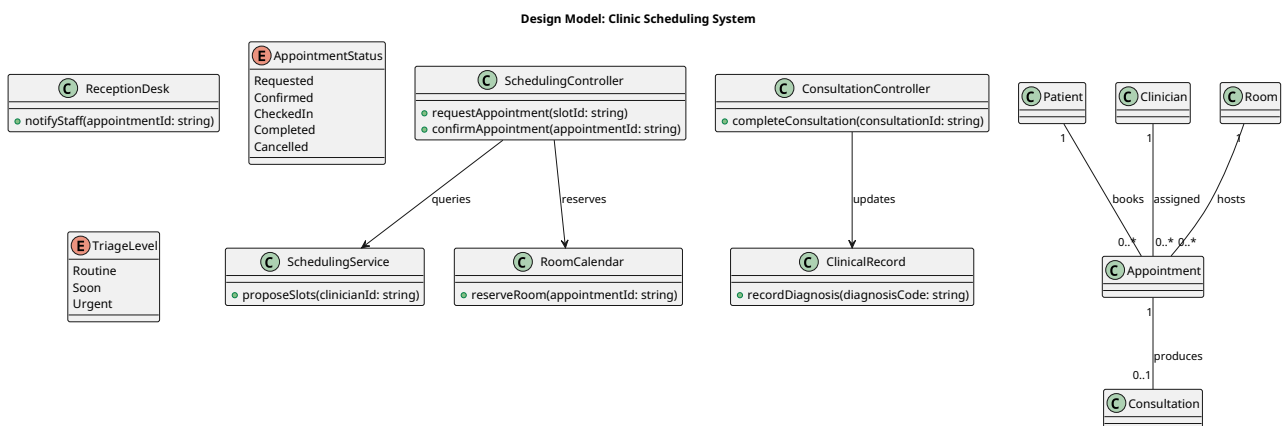
Use Case: CompleteConsultation

Preconditions: - `Appointment.status = CheckedIn`

Postconditions: - `Consultation.status = Completed` - ClinicalRecord is updated

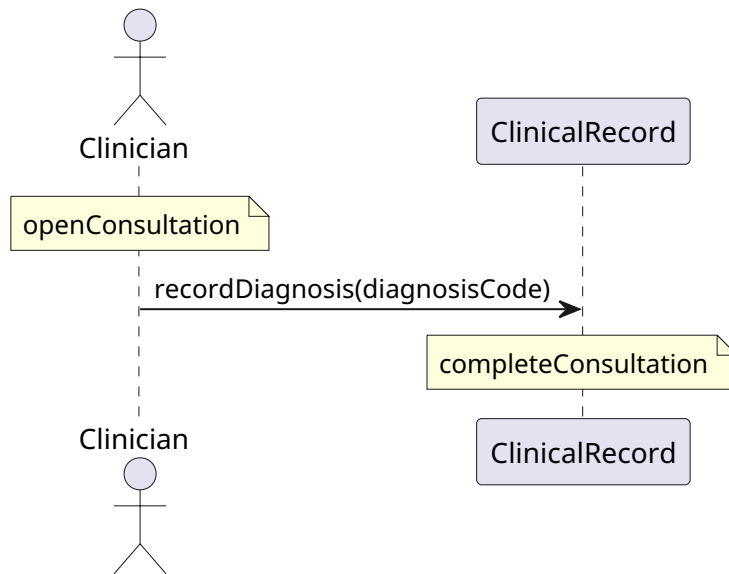
1.3 Design

1.3.1 Structural Views

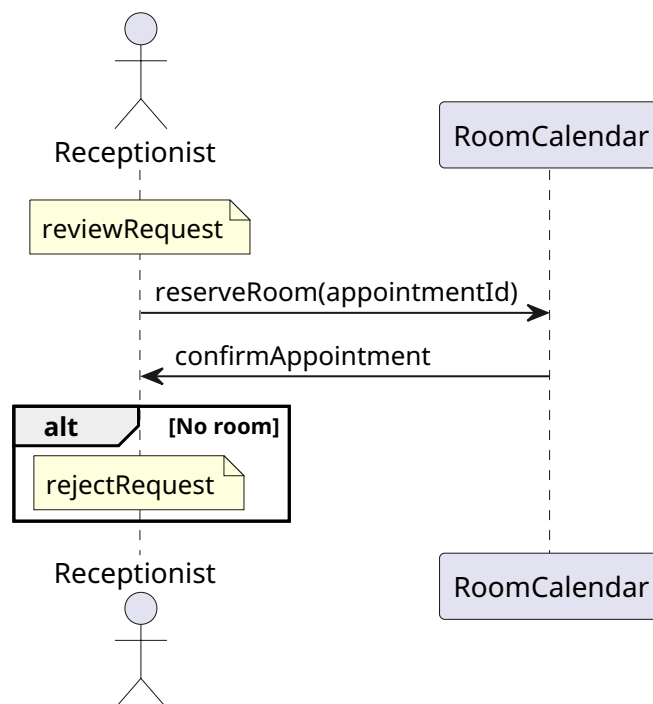


1.3.2 Collaborator Interactions

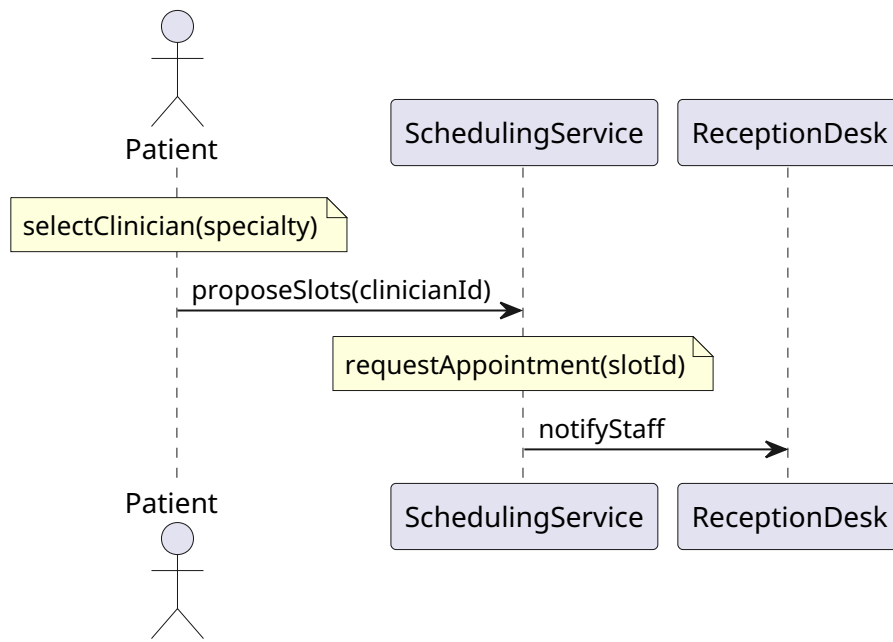
Sequence Diagram: CompleteConsultation



Sequence Diagram: ConfirmAppointment

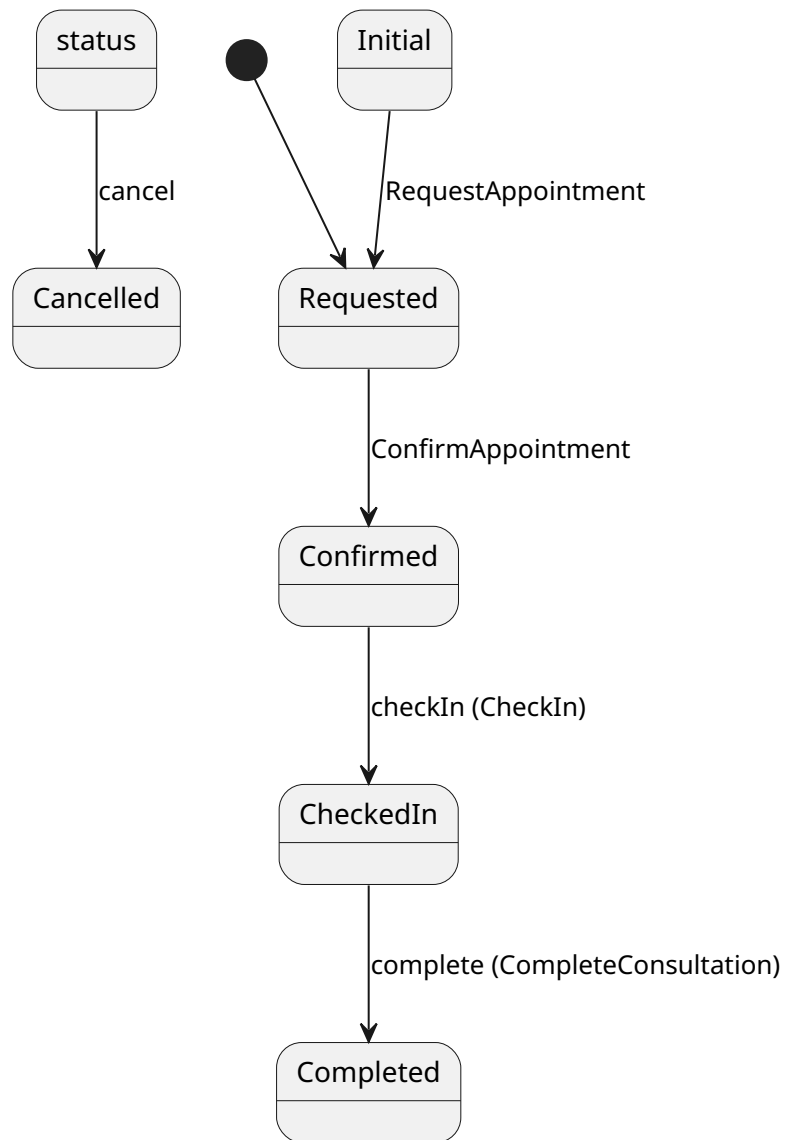


Sequence Diagram: RequestAppointment

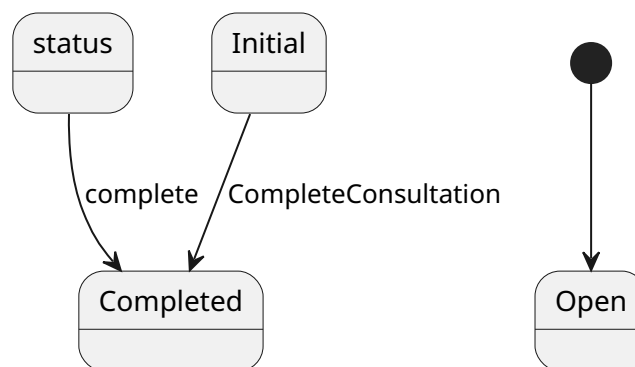


1.3.3 Lifecycle Views

State Machine: Appointment



State Machine: Consultation



1.3.4 Classes

Class	Attributes	Methods	Responsibility Signal
Patient	3	0	3 attributes
Clinician	3	0	3 attributes
Room	2	0	2 attributes
Appointment	3	3	3 operations
Consultation	3	1	1 operations
SchedulingController	0	2	2 operations
ConsultationController	0	1	1 operations
SchedulingService	0	1	1 operations
RoomCalendar	0	1	1 operations
ClinicalRecord	0	1	1 operations
ReceptionDesk	0	1	1 operations

1.3.4.1 CLASS: PATIENT

1.3.4.1.1 Attributes

Attribute	Type	Visibility	Metadata
patientNumber	string	unspecified	-
fullName	string	unspecified	-
email	string	unspecified	-

1.3.4.2 CLASS: CLINICIAN

1.3.4.2.1 Attributes

Attribute	Type	Visibility	Metadata
staffCode	string	unspecified	-
fullName	string	unspecified	-

Attribute	Type	Visibility	Metadata
specialty	string	unspecified	-

1.3.4.3 CLASS: ROOM

1.3.4.3.1 Attributes

Attribute	Type	Visibility	Metadata
roomCode	string	unspecified	-
floor	string	unspecified	-

1.3.4.4 CLASS: APPOINTMENT

1.3.4.4.1 Attributes

Attribute	Type	Visibility	Metadata
status	AppointmentStatus	unspecified	<code>initial(Requested)</code> , <code>state</code>
requestedStart	timestamp	unspecified	-
triageLevel	TriageLevel	unspecified	-

1.3.4.5 CLASS: CONSULTATION

1.3.4.5.1 Attributes

Attribute	Type	Visibility	Metadata
status	string	unspecified	<code>initial(Open)</code> , <code>state</code>
diagnosisCode	string	unspecified	-
completedAt	timestamp	unspecified	-

1.3.4.6 CLASS: SCHEDULINGCONTROLLER

1.3.4.7 CLASS: CONSULTATIONCONTROLLER

1.3.4.8 CLASS: SCHEDULINGSERVICE

1.3.4.9 CLASS: ROOMCALENDAR

1.3.4.10 CLASS: CLINICALRECORD

1.3.4.11 CLASS: RECEPTIONDESK

1.3.5 Enumerations

1.3.5.1 ENUM: APPOINTMENTSTATUS

Values: - Requested - Confirmed - CheckedIn - Completed - Cancelled

1.3.5.2 ENUM: TRIAGELEVEL

Values: - Routine - Soon - Urgent

1.3.6 Relationships

From	Type	To	Label
Patient "1"	–	Appointment "0..*"	books
Clinician "1"	–	Appointment "0..*"	assigned
Room "1"	–	Appointment "0..*"	hosts
Appointment "1"	–	Consultation "0..1"	produces
SchedulingController	→	SchedulingService	queries
SchedulingController	→	RoomCalendar	reserves
ConsultationController	→	ClinicalRecord	updates

1.4 Implementation

1.4.1 Class Operations

1.4.1.1 CLASS: APPOINTMENT OPERATIONS

Method	Visibility	Effects	Metadata	Pre/Post Conditions
checkIn()	unspecified	<code>CheckIn</code> : (Confirmed) status -> CheckedIn	-	-
cancel()	unspecified	<code>Always</code> : status -> Cancelled	-	-
complete()	unspecified	<code>CompleteConsultation</code> : (CheckedIn) status -> Completed	-	-

1.4.1.2 CLASS: CONSULTATION OPERATIONS

Method	Visibility	Effects	Metadata	Pre/Post Conditions
complete()	unspecified	<code>Always</code> : status -> Completed	-	-

1.4.1.3 CLASS: SCHEDULINGCONTROLLER OPERATIONS

Method	Visibility	Effects	Metadata	Pre/Post Conditions
requestAppointment(slotId: string)	+	-	-	-
confirmAppointment(appointmentId: string)	+	-	-	-

1.4.1.4 CLASS: CONSULTATIONCONTROLLER OPERATIONS

Method	Visibility	Effects	Metadata	Pre/Post Conditions
completeConsultation(consultationId: string)	+	-	-	-

1.4.1.5 CLASS: SCHEDULINGSERVICE OPERATIONS

Method	Visibility	Effects	Metadata	Pre/Post Conditions
proposeSlots(clinicianId: string)	+	-	-	-

1.4.1.6 CLASS: ROOMCALENDAR OPERATIONS

Method	Visibility	Effects	Metadata	Pre/Post Conditions
reserveRoom(appointmentId: string)	+	-	-	-

1.4.1.7 CLASS: CLINICALRECORD OPERATIONS

Method	Visibility	Effects	Metadata	Pre/Post Conditions
recordDiagnosis(diagnosisCode: string)	+	-	-	-

1.4.1.8 CLASS: RECEPTIONDESK OPERATIONS

Method	Visibility	Effects	Metadata	Pre/Post Conditions
notifyStaff(appointmentId: string)	+	-	-	-